

P525/2
CHEMISTRY
Paper 2
(Theory)
Nov./Dec. 2025
2½ hours



UGANDA NATIONAL EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

CHEMISTRY

Paper 2
(Theory)

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

*This paper consists of **two** sections; **A** and **B**.*

*Answer **five** questions, including **three** questions from Section **A** and any **two** from Section **B**.*

*Write the answers in the answer booklet(s) provided. Any additional question(s) answered will **not** be marked.*

Begin each question on a fresh page.

A graph paper is provided.

Mathematical tables and silent non-programmable scientific electronic calculators may be used.

Use equations where necessary to illustrate your answers.

Where necessary use the following:

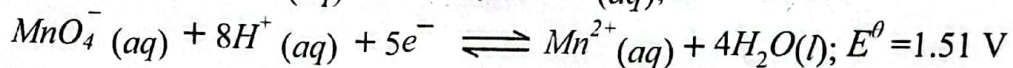
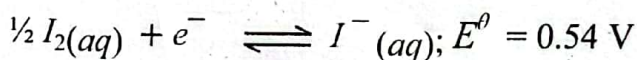
[$H = 1$; $C = 12$; $O = 16$, $Na = 23$]

SECTION A (60 MARKS)

Answer **three** questions from this section.

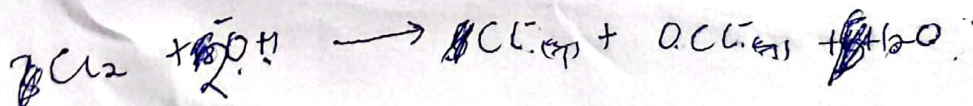
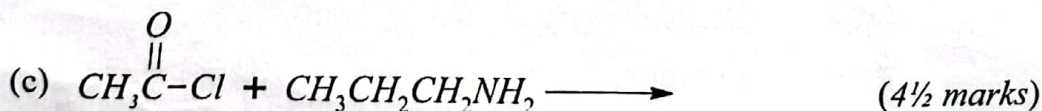
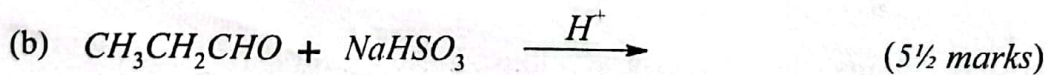
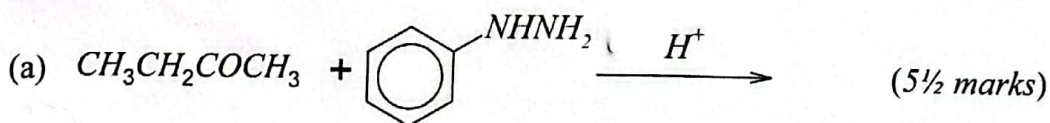
Any additional question(s) answered will **not** be marked.

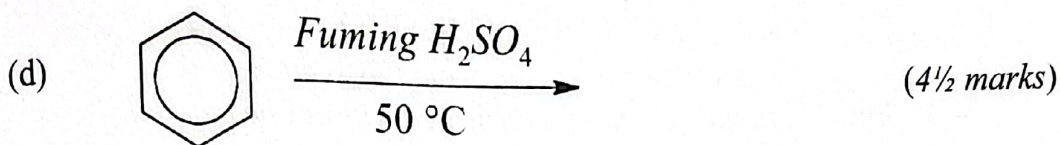
1. (a) (i) Define the term **Standard Electrode potential**. (01 mark)
- (ii) Explain the effect of each of the factors that determine the magnitude of Standard electrode potential. (4½ marks)
- (b) Using a labelled diagram, briefly explain how the Standard Electrode Potential of an iodine electrode can be determined. (05 marks)
- (c) The half-cell equations and the Standard Electrode Potentials for the reaction of iodine and potassium manganate(VII) are given below.



- (i) Write the cell notation for the system and the equation for the overall cell reaction. (2½ marks)
- (ii) Calculate the **e.m.f** for the cell. (1½ marks)
- (iii) Explain why the determination of the concentration of iodine by titration of potassium iodide with potassium manganate(VII) requires no indicators and more accurate results are obtained when the experiment is carried out using acidified manganate(VII) solution. (4½ marks)
- (d) Give **one** application of electrode potentials. (01 mark)

2. Complete the following equations and in each case outline a mechanism for the reaction.





3 (a) Explain the **distribution law**. (05 marks)

(b) (i) Describe how to determine the distribution coefficient of iodine between water and tetrachloromethane. (06 marks)

(ii) 50 cm³ of a 0.1M ammonia solution was shaken with 25 cm³ of trichloromethane at 25 °C. After attaining equilibrium, 12.0 cm³ of the organic layer required exactly 8.0 cm³ of 0.05M hydrochloric acid for complete neutralisation.

Determine the distribution coefficient of ammonia between water and trichloromethane. (06 marks)

(c) State **three** applications of the distribution law. (03 marks)

4. (a) Define the term **melting point**. (01 mark)

(b) Table 1 shows the atomic numbers and melting points of elements in period 3 of the Periodic Table.

Table 1

Elements	Na	Mg	Al	Si	P	S	Cl
Atomic number	11	12	13	14	15	16	17
Melting point (°C)	98	650	660	1410	44	119	-101

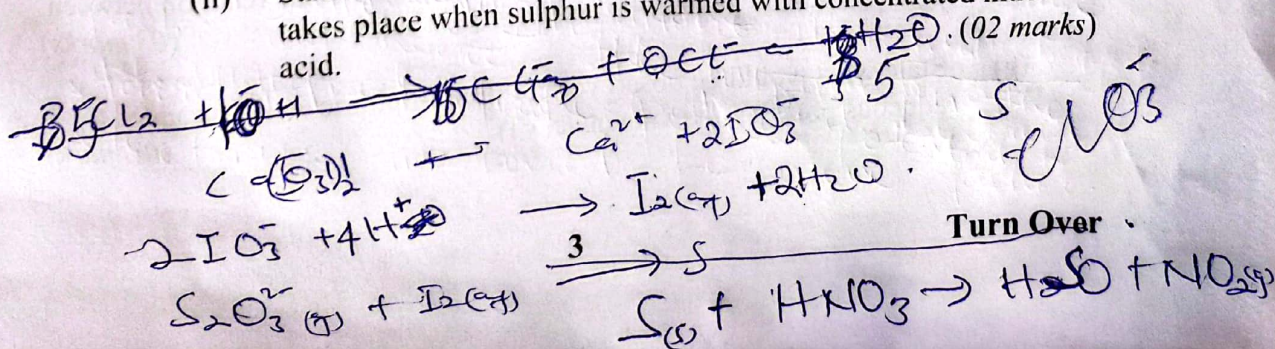
(i) Plot a graph of melting point against atomic number. (04 marks)

(ii) Explain the shape of your graph. (05 marks)

(c) Explain the reactions of Al and Cl with dilute sodium hydroxide solution. (06 marks)

(d) (i) Name **two** crystalline allotropes of sulphur. (02 marks)

(ii) State what is observed and write equations for the reaction that takes place when sulphur is warmed with concentrated nitric acid. (02 marks)



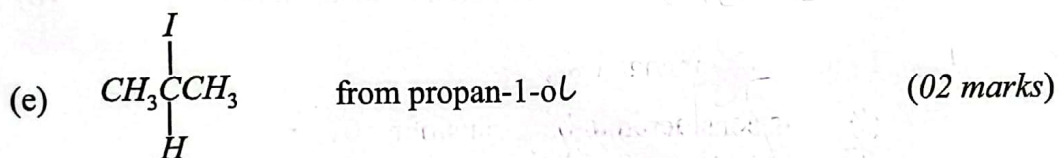
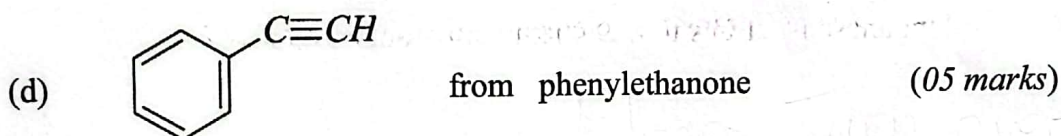
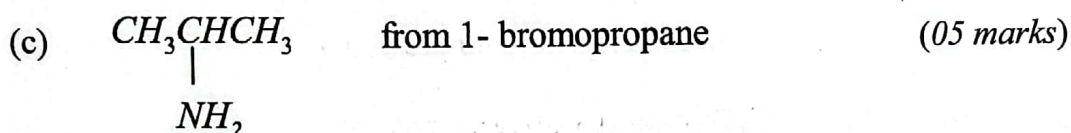
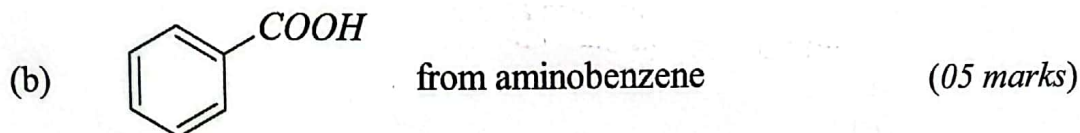
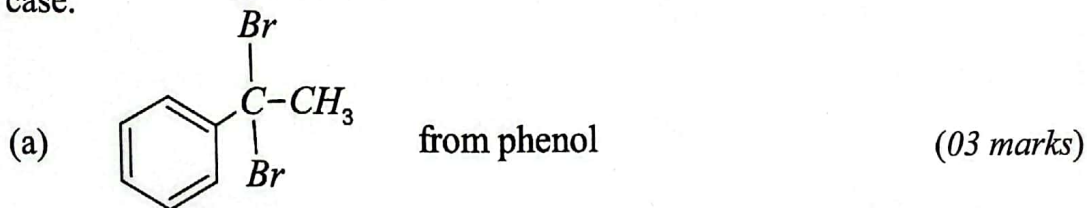
SECTION B (40 MARKS)

Answer **two** questions from this section.

Any additional question(s) answered will **not** be marked.

- 5 (a) Distinguish between a conjugate base of an acid and a conjugate acid of a base and illustrate your answer with suitable examples. (03 marks)
- (b) In the analysis of a solution of a mixture of sodium hydrogen carbonate and sodium carbonate, 20 cm³ of the solution reacted with 4.6 cm³ of 0.1M sulphuric acid using phenolphthalein as an indicator. Another 20 cm³ portion of the same solution, required 14.0 cm³ of the same acid using methylorange as an indicator. Determine the percentage of sodium carbonate in the mixture. (09 marks)
- (c) (i) Sketch the graph of pH against volume of the base for potassium hydroxide and hydrochloric acid titration. (02 marks)
- (ii) Explain the shape of your graph in (c) (i). (4½ marks)
- (d) State **three** applications of acid – base titration. (1½ marks)
6. Manganese is a transition element with atomic number 25 and it is paramagnetic.
- (a) Write the electronic configuration of manganese. (01 mark)
- (b) Explain why manganese;
- (i) is considered to be a transition element. (1½ marks)
- (ii) has variable oxidation states. (02 marks)
- (c) Give a reason why manganese is paramagnetic. (01 mark)
- (d) Describe the reaction of manganese with;
- (i) air. (2½ marks)
- (ii) dilute hydrochloric acid. (03 marks)
- (e) Describe how an aqueous solution of manganese(II) nitrate reacts with aqueous ammonia. (05 marks)
- (f) State what is observed and write the equation for the reaction between potassium manganate(VII) and hydrogen sulphide. (03 marks)
- (g) State what would be observed if dilute sulphuric acid was added to aqueous potassium manganate(VI). (01 mark)

7. Write equations to show how each of the following compounds can be synthesised. Indicate the reagents and conditions for the reactions in each case.



8. Explain each of the following observations and illustrate your answer with equation(s) where necessary.

- Propan-1-ol is a liquid at room temperature whereas propane is a gas under similar conditions. (05 marks)
- The chloride ion has a bigger radius than a chlorine atom. (03 marks)
- When tin(II) chloride solution was added to iodine solution, the brown solution turned colourless. (04 marks)
- When aqueous ammonia is added drop-wise to nickel(II) sulphate solution, a green precipitate is formed which dissolves in excess ammonia forming a purplish blue solution. (04 marks)
- The acid dissociation constant K_a of chloroethanoic acid is $1.3 \times 10^{-3} \text{ mol dm}^{-3}$ but that of ethanoic acid is $1.7 \times 10^{-5} \text{ mol dm}^{-3}$